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Lab Program -01

(a) Write a python program to print " Hello Python"
(b) Write a Python Program to do arithmetical operations.

(a) Program to print " Hello python"

```
print (" Hello python")
```

Output
Hello python

(b) Arithmetical operations

```
def add (P, Q):
 return P+Q
 def subtract (P,Q):
return P-Q
def multiply(P,Q):
return (P*Q)
def divide (P,Q):
return P/Q
print (" Select operation")
print ("a. Add")
print ("b. subtract")
print(" c. Multiply")
print ("d. Divide")
```

```
choice = input("enter your choice:")
num1 = int(input("enter the first no:"))
num2 = int (input("enter the second no:"))
if choice == 'a':
  print (num1,"+", num2, "=", add (num1,num2))
elif choice == 'b':
  print (num1, "-", num2,"=", subtract(num1,num2))
elif choice == 'c':
  print(num1, "*", num2, "=", multiply(num1,num2))
elif choice == 'd':
 print(num1,"/", num2"=", divide (num1,num2))
 else
  print ("invalid choice")
```

Output
Select operation
a. Add
b. subtract
c. multiply
d. divide
enter your choice :d
enter first no: 1
enter second no 2
1/2 = 0.5

Lab Program -02

2) Write a python program to find the area of triangle.

```
base = float(input("enter the base of triangle"))
height = float(input("enter the height of triangle"))

area =(base * height)/2

print("The area of triangle is", area)
```

Output

```
enter the base of triangle 6
enter the height of triangle 2
area of triangle is 6

enter the base of triangle 7
enter the height of triangle 2
area of triangle is 7
```

Lab Program - 03

③ Solve quadratic equations

```
import cmath
a = float(input('enter a :'))
b = float(input('enter b:'))
c = float(input('enter c:'))

#calculate the discriminant
d = (b**2) - (4*a*c)

# find two solutions
sol1 = (-b-cmath.sqrt(d)) /(2*a)
sol2 = (-b+cmath.sqrt(d)) /(2*a)
print("The solution are {0} and {1}'.
    format(sol1,sol2))
```

Output

```
enter a :3
enter b :5
enter c :9
Solutions are (-0.83333333333-1.
5184055965 2405j) and-0.83333
33333334+ 5184055965224 05j
```

Lab Program - 04

4) Write a python program to swap two numbers

```
P = int (input ("enter value for P:"))
Q = int (input ("enter value for Q:"))
  temp = P
    P = Q
    Q = temp
print ("The value of P after swapping:", P)
print ("The value of Q ater swapping:", Q)
```

Output

```
enter value P: 12
enter value Q: 43
The value of P after swapping
43
The value of Q alter swapping
12
```

Lab Program 05

(5) write a python program to celsius to celsius to Faranheit

```
# convert celsius to Fahrenheit
celsius = float (input ("temperature value in degree
                celsius: "))

fahrenheit = (celsius *1.8) + 32
print (" the %.2f degree celsius is equal to : %.2f
        fahrenheit " %(celsius , fahrenheit))

# convert fahrenheit to celsius

fahrenheit = float (input (" temperature value in
            degree fahrenheit: "))

celsius = (fahrenheit - 32) / 1.8

print (" the %.2f degree fahrenheit is equal to:
        %.2f celsius" %(fahrenheit, celsius))
```

Output

```
temperature value in degree celsius : 32
the 32.00 degree celsius is equal to : 89.60
                                  fahrenheit

temperature value in degree fahrenheit :94

The 94.00 degree fahrenheit is equal to 34.44
                                  celsius
```

Lab Program - 06

⑥ Write a python Program to check if numbers is odd or Even

```
number = int(input("enter a number"))
if number % 2 == 0:
        print("{0} is even".format(number))

else:
        print("{0} is odd".format(number))
```

Output

```
Enter a number 2

2 is even
```

Lab Program -07.

(7) Write a python program to print all Prime numbers in an interval.

```
lowervalue = int(input("enter lowervalue"))
uppervalue = int(input("enter uppervalue"))
print("prime numbers")
for number in range (lower value, uppervalue+1)
    if number >1:
        for i in range(2,number):
            if (number%i) = = 0 :
                break
        else:
            print(number)
```

Output

```
enter lower value 6
enter upper value 10
Prime numbers    7
```

Lab Program -08

8) write a python program to find the factorial of a number.

```
num = int(input("enter a number"))
fact=1
if num<0:
    print("the factorial doesnot exist for negative
                    numbers")
elif num==0:
    print(" the factorial of 0 is 1")
else:
    for i in range (1, num+1):
        fact = fact *i
    print(" The factorial of given no is ", fact)
```

output

enter a number 5

The factorial of given number is 120

Lab Program -09

(9) Write a python program to display the multiplication table.

```
number =int (input ("enter the number to print the
                multiplication table:"))

count=1

#while loop for iterating the multiplication 10 times
print ("The multiplication Table of: ", number)

while count<=10:
    number = number *1
    print (number *count)
    count +=1
```

Output

```
Enter the number to print the multiplication
table :10

The multiplication Table of :10
 10
 20
 30
 40
 50
 60
 70
 80
 90
```

Lab Program -10

(10) Write a python program to multiply two matrices

```
import numpy as np
#Define two matrices
A= np.array([1,2],
            [3,4]])

B = np.array([[[5,6]],
        [7,8]])
#Multiply the matrices
result = np.dot(A,B)

#Print the result
print("Matrix A:")
print(A)

print("Matrix B:")
print(B)

print("Product A and B:")
print(result)
```

Output

```
Matrix A:
[[1 2]
 [3,4]]

Matrix B:
[[5 6]
 [7 8]]
Product A and B:
[[19 22]
 [43 50]]
```

Lab Program - 11

11) write a python Program to Find LCM & GCD using function

```
def g(a,b):
    gcd = 1
    for i in range (1,a+1):
        if a%i == 0 and b%i==0:
            gcd=i
    return gcd.

first = int(input("enter the first number:"))
second = int(input("enter second number:"))
print("GCD of %d and %d"%(first,second,lcm))
```

Output
enter first number: 3
enter second number: 4

HCF or GCD of 3 and 4 is 1

LCM of 3 and 4 is 12

Lab Program -12

12) Write a python program to read a word and print the number of letters, vowels in the word

```
string = input("enter a string:")
vowels = ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']
tot1 = 0
tot2 = 0
for line in string:
  print(line)
for char in string:
    if char >= 'a' and char <= 'z':
    if char not in vowels:
        tot2 = tot2+1
   elif char >= 'A' and char <= 'Z':
      if char not in vowels:
         tot2 = tot2+1
print ("Number of consonant are:")
print (tot2)
for char in string:
     if char in vowels
         tot1 = tot1+1
print ("Number of vowels are:")
print(tot1)
```

Output

Enter a string : Hello world

H
e
l
l
o
w
o
r
l
d
Number of consonant are :
7
Number of vowels are :
3

Lab Program -13

(13) Write a python program to input an array of n numbers and find seperately the sum of positive numbers and negative numbers

```
def sep(numbers):
    pos = 0
    neg = 0
    for num in numbers:
        if num>0:
            pos+ =num
        elif num<0:
            neg+ =num
    return pos,neg
n = int(input("enter the length of array"))
numbers = []
for i in range(n):
    num = int(input("enter the number"))
    numbers.append(num)
pos, neg = sep(numbers)
print("sum of positive number is", pos)
print("sum of negative number is", neg)
```

Output

```
enter the length of array 2
enter the number 2
enter the number 5
Sum of positive number 7
sum of negative number 0
```

Lab Program -14

(14) Write a python program to search an element using linear equation implement a sequential search.

```
def linear-search(alist, key):
    for i in range(len(alist)):
        if alist[i] == key:
            return i
    return -1

alist = input("enter the list of numbers:")
alist = alist.split()
alist = [int(x) for x in alist]
key = int(input('The number to search for:'))
index = linear-search(alist, key)
if index < 0:
    print('{} was not found.'.format(key))
else:
    print('{} was found at index {}.'.format(
                          key, index))
```

Output

```
Enter the list of numbers 6 7 8 9 5 4 3 9 9
The number to be search for: 3
3 was found at index 6
```

Lab Program -15

(15) Write a python program to search an element using binary search

```
def binary-search (arr, target):

    low=0
    high= len(arr)-1

    while low <= high:
        mid =(low+ high)//2
        if arr[mid ]= =target:
            return mid         # Element found at index mid
        elif arr [mid]< target:
            low =mid+1
        else:
            high= mid-1

    return -1        # Element not found
arr =[1,3,5,7,9,11,13,15]
target =int (input ("enter the element to search :"))
result = binary-search(arr,target)

if result != -1:
    print ("Element found at index :",result)

else:
    print ("Element not found in the list.")
```

Lab Program - 15

Output

enter the element to search 9

Element is found at 4 index

Output

enter the element to search 18

element not found in the list

Lab Program -16

(16) Write a python program to insert a number in sorted array

```
del insert(list,n):
    list. append (n)
    sorted-list = sorted (list )
    return sorted-list
    list =[1, 2, 3 ]
    n = 3
    print (insert (list,n))
```

Output

[1, 2, 3,3]

Lab Program -17

(17) Write a python program to stimulate stack operation.

```
stack = []
stack.append('a')
stack.append('b')
stack.append('c')
print('initial stack')
print(stack)
print('element deleted from stack')
print(stack.pop())
print('after stack.pop())
print('after element are popped')
print(stack)
```

Output

```
initial stack
['a', 'b', 'c']
element deleted from stack
c
b
a
after element are popped
```

Lab Program -18

(18) Write a python program to stimulate stack operation to draw shapes & GUI controls

```
from tkinter import*
from functools import partial

def validateLogin(username, password);
    print("username entered:", username.get())
    print("password entered:", password.get())
    return

t= Tk()
t.geometry('400x150')
t.title('Tkinter Login Page')
usernameLabel = Label(t, text = "User Name").grid
                    (row=0, column=0)

username = String var()
usernameEntry= Entry(t, textvariable=username).grid
                    (row= 0, column=1)
passwordLabel=Label(t, text ="Password"). grid(
                    row=1, column=0)

password = Stringvar()
passwordEntry = Entry(t, textvariable=password, show='*')
                    .grid (row=1, column=1)

validatelogin = partial (validateLogin, username,
                    password)
```

Lab Program -18

```
login_Button = Button(t, text="login", command=validate
        Login).grid(row=4, column=0)

t.mainloop()
```

Output

Username entered : Alice

Password entered : 1234

Lab Program -19

(19) Write a python programs to using the built in Methods of the string, list, dictionary

Built in methods of the string

```
a='hi'
a.capitalize()
>>>> 'Hi'

a='hi'
a.upper()
>>> 'HI'

a='HI'
a.lower()
>>> 'hi'

a='hi'
a.title()
>>> 'Hi'

a = 'hi'
a.swapcase()
>>>> 'HI'

a = 'Happy Birthday'
a.split()

>>> ['Happy', 'Birthday']
```

a='Happy Birthday'
a.center(18,'*')
>>>>'** Happy Birthday**'

a ='Happy Birthday'
a.count('Birthday')
>>>> 1

a='Happy Birthday'
a.replace('Happy','wishyou happy')
>>> 'wishyou happy Birthday'

a= 'Happy Birthday'
b="happy"
a.join(b)
>>>'hHappy BirthdayaHappy BirthdaypHappy BirthdaypHappy Birthdayy'

Built in methods in the list

Numbers = [2,4,6]
Numbers.append (3)
print ("after appending", Numbers)

>>> [2,4,6,3]

```
Languages = ['kannada', 'English']
Lang = ['Tamil, Hindi']
Languages.extend (Lang)
print('Languages list is', Languages)
>>>> ['kannada', 'English', 'Tamil', 'Hindi']

Lang = ['Kannada' 'English']
Lang.insert (2, Hindi')
print ('Language is', Lang)
>>> ['kannada', 'English', 'Hindi']

Lang = ['python', 'java']
Lang.pop(1)
print (Lang)
>>> ['python']

cities = ["oslo", "delhi", "washington", "london",
          'seattle", "paris", "washington"]

cities.count('seattle')
>>>> 1
```

Built in methods of dictionary

```
a = {'m', 'b', 'v'}
n = 1
d = dict.fromkeys(a, n)
print(d)
>>>>{'b': 1, 'm': 1, 'v': 1}
```

```
Names = { 1: 'Abhi', 2: 'Raja', 3: 'Rakesh'}
Names.pop ('1')
print (Names)
>>>{2: 'Raja', 3: 'Rakesh'}
```

```
Names = { 1: 'Abhi', 2: 'Raja', 3: 'Rakesh'}
Names.clear()
print (Names)
>>>> {}
```

```
Names = { 1: 'Abhi', 2: 'Raja', 3: 'Rakesh'}
Names.get ('2')
print (Names)
>>> Raja
```

```
Numbers = {'w': 4, 'e': 2, 'r': 12, 't': 9}
sorted-no = sorted (Numbers)
print (sorted-no)
>>> 'e', 'r', 't', 'w'
```

Lab Program-20.

(20) Write a python program to demonstrate exception handling

```
try:
    temp = [1,2,3]
    temp[3]
except Exception as e:
        print("exception block handling:",e)

else:
    print ("No exception found")

finally:
    print ("finally block executed")
```

Output
Exception block handling
finally block executed

Seen 6/6/25